



Bose SoundLink Air

Bose recently announced its first Apple AirPlay device - the SoundLink Air which allows you to play music from your Apple devices over WiFi



Speaker-fitted bed?

Time4sleep has come out with a Sound leather bed that has a speaker section fitted into the head-rest.



a normal speaker unit along with an RF transmitter and receiver and antennas. The stereo system has a transmitter which will convert electrical signal to radio waves. These radio waves are broadcast by the antenna on the stereo system, which are received by the antenna and receiver on the wireless speaker. This signal is then converted back to electricity. An amplifier boosts the power of the signal thereby driving the speaker. Nowadays a lot of speakers incorporate bluetooth technology, where as long as your speaker or headphone is seen as a bluetooth device, it can be paired to play music off any bluetooth enabled device such as your phone or tablet. While they help in reducing that menace called wires you will still need to power these speakers by either connecting them to a power source or using batteries. The sound quality may not be audiophile-grade, but you still have the convenience of setting up the speakers anywhere.

4 FLAT-PANEL SPEAKERS

With our increasing affinity towards slim-everything and minimalist design, it is only natural for bulky speakers to make way for something more aesthetic and which just disappears in the background. Enter flat panel speakers, which can be hung on walls or



made in neutral colours which make them less conspicuous. It is difficult to make electrodynamic flat panel speakers for the simple reason that it is difficult to vibrate the entire diaphragm evenly, while being

placed so close to the coil. As a result the frequency response is not so great. But there are work-arounds.

Voice coils are mounted on flat panels which act as sound sources. Light materials such as styrofoam are used to make flat panel speakers.

5 ELECTROSTATIC SPEAKERS

This is completely different in principle from an electrodynamic speaker, in the sense that it uses a high voltage electric field instead of magnetic field to



drive a statically charged thin membrane. This thin membrane is the diaphragm that pushes the air molecules to produce sound. It is surrounded by metal panels (stators) that receive the audio signal from the amplifier. There is a small gap between the stators and the membrane which is covered by spars which prevents the membrane from getting in contact with the stators. The diaphragm (membrane) get its charge from the power supply which charges it positively. It stays constantly biased that way. Whereas the stators are connected to the amplifier. The contacts from the amplifier pass through a step-up transformer and connect to the stators. This causes the stators to be oppositely charged. The diaphragm gets pulled towards the negatively charged stator whereas the positively charged stator pushes it. As the sound-waves pass through the amplifier, the polarity on the stators keeps changing,

thereby causing a constant back and forth motion of the diaphragm which ultimately produces the sound.

Apart from being flat in nature and aesthetically pleasing, this speaker has a uniform force across the full surface of the driver. It is much lighter than a traditional electrodynamic speaker provided the air displacement is the same in either case. Another important factor is that there is fairly little lag between the original audio signal and the audio output. In electrodynamic speakers, the pronounced lag can make distortions very prominent.

6 PIEZOELECTRIC SPEAKERS

Piezoelectric speakers as the name suggests work on the piezoelectric effect: which is basically that charge accumulates in certain materials such as crystals, ceramics, etc., based on the mechanical stress applied. In a piezoelectric speaker, there is a thin polarised ceramic plate which is attached to a diaphragm film. When voltage is applied to the ceramic plate, it starts vibrating thus converting electrical energy into mechanical stress thereby producing the sound. They are most commonly found as beepers in your electronic devices or as tweeters in computer speakers. They have a threshold for higher frequencies and are resistant to loads that would harm voice coils in most of the high frequency drivers. They are also comparatively cheaper and do not take up much space.

One of the major disadvantages is that its frequency response is not that great as compared to traditional speakers, which is



the main reason why it is used in non-critical single frequency applications. They can also be used in sonar applications, where they act as sound transmitters as well as receivers. **[1]**